

Name: **Nathan Mull**

BUID: **12345678**

Workshop 1: Matrix-Vector Equations

1. *Review.* Determine if the vector $\begin{bmatrix} 9 \\ 3 \\ -14 \end{bmatrix}$ is in the span of $\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix} \right\}$.
2. Calculate the matrix-vector multiplication $\begin{bmatrix} 1 & -4 & 3 \\ -6 & 2 & 5 \end{bmatrix} \begin{bmatrix} -5 \\ -1 \\ -1 \end{bmatrix}$.
3. Write a vector equation and a linear system with the same solution set as the matrix-equation $\begin{bmatrix} -4 & 5 & 5 \\ 0 & 1 & 1 \\ -9 & 3 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$.
4. Determine if the equation $\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ -2 & -1 & -4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 9 \\ 3 \\ -14 \end{bmatrix}$ has a solution. If it does, then determine a general form solution.
5. Determine if the equation $\begin{bmatrix} 1 & 3 & -1 \\ 1 & -1 & 3 \\ -1 & 2 & -4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 10 \\ -2 \\ 5 \end{bmatrix}$ has a solution. If it does, then determine a general form solution.
6. Determine the values of b_1 and b_2 such that the RREF of $\begin{bmatrix} 1 & -1 & b_1 \\ -2 & 2 & b_2 \end{bmatrix}$ ~~has~~ ^{does not have} the row $[0 \ 0 \ 1]$.
7. Determine the values of b_1 , b_2 , and b_3 such that the RREF of $\begin{bmatrix} 1 & -1 & b_1 \\ 2 & -1 & b_2 \\ 0 & -1 & b_3 \end{bmatrix}$ ~~has~~ ^{does not have} the row $[0 \ 0 \ 1]$.
8. Determine if $\begin{bmatrix} -4 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} 8 \\ -10 \end{bmatrix}$ span all of $\mathbb{R}^2$. Also determine if $\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 19 \\ 11 \\ 0 \end{bmatrix}$ span all of $\mathbb{R}^3$. Explain your answers.
9. Determine a matrix $A \in \mathbb{R}^{2 \times 3}$ ^{with distinct entries} such that the equation $A\mathbf{x} = \mathbf{0}$ has $\mathbf{1}$ (the all-ones vector) as a solution.

Write your answers below and on the back of this page.

1. $\begin{bmatrix} 1 & 1 & 3 & 9 \\ 0 & 1 & 2 & 3 \\ -2 & -1 & -4 & -14 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 & 9 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 & 9 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ **NO**

2. $\begin{bmatrix} 1 & -4 & 3 \\ -6 & 2 & 5 \end{bmatrix} \begin{bmatrix} -5 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1(-5) - 4(-1) + 3(-1) \\ -6(-5) + 2(-1) + 5(-1) \end{bmatrix} = \begin{bmatrix} -4 \\ 23 \end{bmatrix}$

3. $x_1 \begin{bmatrix} -4 \\ 0 \\ -9 \end{bmatrix} + x_2 \begin{bmatrix} 5 \\ 1 \\ 3 \end{bmatrix} + x_3 \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$
 $-4x_1 + 5x_2 + 5x_3 = 1$
 $x_2 + x_3 = -1$
 $-9x_1 + 3x_2 + x_3 = 2$

4. $\begin{bmatrix} 1 & 1 & 3 & 9 \\ 0 & 1 & 2 & 3 \\ -2 & -1 & -4 & -14 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 3 & 9 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 6 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ **NO**

5. $\begin{bmatrix} 1 & 3 & -1 & 10 \\ 1 & -1 & 3 & -2 \\ -1 & 2 & -4 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -1 & 10 \\ 0 & -4 & 4 & -12 \\ 0 & 5 & -5 & 15 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -1 & 10 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$x_1 = 1 - 2x_3$
 $x_2 = 3 + x_3$
 x_3 is free

9. $\begin{bmatrix} 6 & 1 & -5 \\ 3 & 1 & -4 \end{bmatrix} \vec{x} = \vec{0}$

6. $\begin{bmatrix} 1 & -1 & b_1 \\ -2 & 2 & b_2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & b_1 \\ 0 & 0 & b_2 + 2b_1 \end{bmatrix}$
 any b_1, b_2 s.t. $2b_1 + b_2 = 0$, eg. $b_1 = 1, b_2 = -2$

7. $\begin{bmatrix} 1 & -1 & b_1 \\ 2 & -1 & b_2 \\ 0 & -1 & b_3 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & b_1 \\ 0 & 1 & b_2 - 2b_1 \\ 0 & -1 & b_3 + b_1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & b_1 \\ 0 & 1 & b_2 - 2b_1 \\ 0 & 0 & b_3 + b_1 - 2b_1 \end{bmatrix}$
 $-2b_1 + b_2 + b_3 = 0$

8. $\text{span} \left\{ \begin{bmatrix} -4 \\ 5 \end{bmatrix}, \begin{bmatrix} 8 \\ -10 \end{bmatrix} \right\} \neq \mathbb{R}^2$, $v \in \text{span} \{ \dots \} \Rightarrow v = c \begin{bmatrix} -4 \\ 5 \end{bmatrix}$
 so, e.g., $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \notin \text{span} \{ \dots \}$

$\text{span} \left\{ \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 19 \\ 11 \\ 0 \end{bmatrix} \right\} \neq \mathbb{R}^3$ $\begin{bmatrix} 2 & 19 \\ 0 & 11 \\ 1 & 0 \end{bmatrix}$ has

at most 2 pivots, needs 3 to span $\mathbb{R}^3$

Continue your work here.